

NATIONAL COLLEGE OF IRELAND
SCHOOL OF COMPUTING

Master of Science in Data Analytics (MSCDAD)

SUPERVISION & WEEKLY PROGRESS REPORT (CA2 / PRACTICUM)

Student Name:	Rohitkumar Amritlal Jaiswal	Student ID Number:	x25119613
Academic Supervisor:	Dr. Thanos Staikopoulos	Reporting Period:	Week 1 (Kickoff) 29 Sep 2026
Project Title:	IrishLogix: Coupling Port Congestion & Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation for Post-Brexit Irish Freight Logistics	Meeting Mode & Time:	Formal Review (45 Mins)

1. ACTIVITIES COMPLETED THIS WEEK (WHAT?)

NCI Reflective Prompt: Reflect on what has happened in your project this week. Detail concrete technical, analytical, or literature tasks completed.

- Proposal Finalisation (CA2):** Refined and compiled the 4,691-word Research in Computing Proposal PDF incorporating CA1 supervisory feedback. Streamlined the architecture from eight overly broad modules down to three tightly coupled core components: (C1) Port Congestion Classifier, (C2) Customs Delay Regressor, and (C3) Carbon-Aware Pareto Route Optimizer.
- Literature Review Reference Archiving:** Curated, validated, and archived all 7 foundational peer-reviewed papers and EU legislative instruments (*Directive (EU) 2022/2464 (CSRD)*, *Directive (EU) 2023/959 (ETS2)*, *Regulation (EC) No 561/2006*, Cheng et al. 2024, Zamani et al. 2022, etc.) into Literature_Review_References.zip.
- Ethics Consideration Submission:** Completed and signed the official NCI School of Computing Declaration of Ethics Consideration Form, verifying no human participants and documenting all open public secondary data sources (CSO, EMODnet AIS, OpenStreetMap) and synthetic data generators.
- Project Pitch Script Formulation:** Structured a strict 5-stage verbal pitch (*Problem → Gap → Idea → Method → Contribution*) and anticipated technical defense responses for the scheduled 1 October panel.

2. EVALUATION & CRITICAL REFLECTION (SO WHAT?)

NCI Reflective Prompt: Consider what that meant for your project progress. What were your successes? What challenges or bottlenecks still remain?

- Scope Streamlining & Feasibility:** The reflection on CA1 feedback proved that the initial 8-module scope was too expansive for a single-semester MSc Practicum. Pruning driver tachograph and telematics modules successfully eliminated GDPR and privacy risks while establishing a realistic, rigorous technical scope.
- Technical Risk Assessment & Synthetic Data:** Identified that shipment-level customs clearance durations across Irish sea corridors are commercially confidential and strictly restricted by Irish Revenue. Rather than relying on guesswork, creating a calibrated synthetic dataset based on aggregate CSO trade statistics provides a reproducible, ethically compliant approach.
- Methodological Alignment:** Confirmed that Gradient Boosting (XGBoost) is optimal for tabular port and customs delay data, while bi-objective Pareto optimization (minimizing operational cost vs. carbon intensity) accurately captures the economic and environmental trade-offs required under CSRD and ETS2.

3. ACTION PLAN & NEXT STEPS (NOW WHAT?)

NCI Reflective Prompt: What concrete steps will you take next to address challenges and progress toward the next milestone?

- **Oral Pitch Delivery (1 October):** Deliver the 1–2 minute project idea pitch to the panel and technical peers, adhering to the timed 100-second delivery structure.
- **Supervisor Consultation:** Discuss and align on refined Primary Research Questions, Sub-Questions, and SMART Objectives with Dr. Thanos Staikopoulos during the supervisory review.
- **Data Pipeline Ingestion:** Begin Phase 1 data ingestion: download CSO Port Traffic tables, extract EMODnet vessel density data, and construct the Irish corridor road network graph via OpenStreetMap.
- **Baseline Simulation Model:** Develop the FIFO shortest-route spreadsheet simulation baseline planner against which the machine learning and route optimization model will be benchmarked.

4. SUPERVISOR MEETING LOG & AGREED ACTIONS

Agenda Items Discussed:	1. CA2 Proposal Submission & Literature Reference Package. 2. Presentation strategy for the 1 October Project Pitch. 3. Updated 15-week Gantt Chart milestones and deliverable timeline.
Supervisor Guidance & Feedback:	• Ensure the pitch clearly articulates the gap between prediction and optimization. • Maintain strict focus on the 3 core components (C1, C2, C3). • Prepare for Q&A regarding synthetic customs data calibration.
Agreed Action Items & Deadlines:	1. Deliver 1–2 minute pitch on 1 October 2026. 2. Refine Research Title, RQ, and SMART Objectives for next meeting. 3. Submit weekly progress updates every Tuesday.
Target Date for Next Meeting:	Week 2 Scheduled Supervision Catch-up (Tuesday, 6 October 2026)

5. PROJECT MILESTONE & GANTT TRACKING

Milestone	Target Week	Current Status	Deliverable Summary
Milestone 1 (M1)	Week 3	IN PROGRESS	Baseline FIFO model + CSO/EMODnet data pipeline complete
Milestone 2 (M2)	Week 9	PLANNED	XGBoost models (C1, C2) coupled with Pareto Optimizer (C3)
Milestone 3 (M3)	Week 13	PLANNED	Dual-system comparative evaluation & Streamlit UI dashboard
Milestone 4 (M4)	Week 15	PLANNED	Final MSc Dissertation, code repository, and Viva Voce

6. FORMAL DECLARATION & SIGN-OFF

Student Signature: Rohitkumar Amritlal Jaiswal Date: 29 September 2026	Supervisor Signature: _____ Date: _____
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